D-FINE BASED INTELLIGENT SURVEILLANCE VIDEO COMPRESSION USING ADAPTIVE MOTION ESTIMATION

A PROJECT REPORT

Submitted by

AFSHANA R

DHARINI N

KIRUBHA SHRIVAISHNAVI G

MEERA B

in partial fulfilment for the award of the degree of

BACHELOR OF ENGINEERING

in

ELECTRONICS AND COMMUNICATION ENGINEERING

K. RAMAKRISHNAN COLLEGE OF ENGINEERING

(AUTONOMOUS)

SAMAYAPURAM, TRICHY

ANNA UNIVERSITY

CHENNAI 600 025

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Submitted by

AFSHANA R	**(8115U23EC002)**
DHARINI N	**(8115U23EC021)**
KIRUBHA SHRIVAISHNAVI G	**(8115U23EC051)**
MEERA B	**(8115U23EC062)**

in partial fulfilment for the award of the degree of

BACHELOR OF ENGINEERING

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Under the Guidance of

Dr. K. PRIYADARSHINI, Ph.D.,

Professor

Department of Electronics and Communication Engineering

K. Ramakrishnan College of Engineering (Autonomous)

K. RAMAKRISHNAN COLLEGE OF ENGINEERING

(AUTONOMOUS)

Under

ANNA UNIVERSITY, CHENNAI

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K. RAMAKRISHNAN COLLEGE OF ENGINEERING
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Under
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BONAFIDE CERTIFICATE

Certified that this project report titled **"D-FINE-BASED INTELLIGENT SURVEILLANCE VIDEO COMPRESSION USING ADAPTIVE MOTION ESTIMATION"** is the bonafide work of **AFSHANA R (8115U23EC002), DHARINI N (8115U23EC021), KIRUBHA SHRIVAISHNAVI G (8115U23EC051), MEERA B (8115U23EC062)** who carried out the project work under my supervision. Certified further, that to the best of my knowledge the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

SIGNATURE

Dr. M. MAHESWARI, M.Tech., Ph.D.,
HEAD OF THE DEPARTMENT
Professor,
Department of Electronics and Communication Engineering,
K. Ramakrishnan College of Engineering (Autonomous),
Samayapuram – 621 112.

SIGNATURE

Dr. K. PRIYADARSHINI, Ph.D.,
SUPERVISOR
Professor,
Department of Electronics and Communication Engineering,
K. Ramakrishnan College of Engineering (Autonomous),
Samayapuram – 621 112.

Submitted for the End Semester Examination held on …….

SIGNATURE OF INTERNAL EXAMINER

NAME:

DATE:

SIGNATURE OF EXTERNAL EXAMINER

NAME:

DATE:



K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)
Under
ANNA UNIVERSITY, CHENNAI



DECLARATION BY THE CANDIDATES

We declare that to the best of our knowledge the work reported herein has been composed solely by ourselves and that it has not been in whole or in part in any previous application for a degree.

Submitted for the project Viva-Voce held at K. Ramakrishnan College of Engineering on _____.

_____

SIGNATURE OF THE CANDIDATES

AFSHANA R (8115U23EC002)
DHARINI N (8115U23EC021)
KIRUBHA SHRIVAISHNAVI G (8115U23EC051)
MEERA B (8115U23EC062)

K. RAMAKRISHNAN COLLEGE OF ENGINEERING (AUTONOMOUS)
SAMAYAPURAM, TRICHY – 621 112

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.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	**3**	**2**	**3**	**2**	**3**	**2**	**2**	**3**	**3**	**2**	**2**	**2**	**3**	**3**



K.RAMAKRISHNAN COLLEGE OF ENGINEERING
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ABSTRACT

Video surveillance systems operate continuously and generate enormous volumes of data, which places heavy demands on storage and network bandwidth. Conventional codecs such as H.264 and H.265 apply the same motion estimation and encoding strategy to every region of a frame, so a large share of the bit budget is spent on static background that carries little useful information. This work presents an intelligent surveillance video compression framework that integrates the D-FINE object detector with adaptive motion estimation. D-FINE, a real-time DETR-based detector, identifies objects of interest such as persons and vehicles, and the detected bounding boxes are converted into a Region of Interest (ROI) mask for every frame.

Based on this mask, object regions are processed with small block sizes and a wide search range to preserve detail, while background regions use larger blocks and a narrower search range to achieve higher compression with lower computation. The system is implemented in Python using Py Torch, OpenCV and FF mpeg. It is evaluated on four surveillance videos and compared with an existing YOLOv8-based method using size reduction, PSNR and SSIM. The existing method reduces the size of only one video, by 5.05%, and increases the size of the other three, whereas the proposed D-FINE method reduces the size of every video by 80.23% to 87.56%, with an average size reduction of 82.879%. The proposed method also gives a higher PSNR for all four videos, with an average of 34.49 dB against 33.30 dB for the existing method, and an average SSIM of 0.9651. These results confirm that object-aware, content-adaptive compression greatly reduces storage and bandwidth requirements while maintaining the visual quality of important regions, which makes the framework suitable for modern surveillance systems.

Keywords: D-FINE, Surveillance Video, Video Compression, Adaptive Motion Estimation, Region of Interest, Object Detection

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CHAPTER-1
INTRODUCTION

1.1. INTRODUCTION

Video surveillance has become an essential part of modern infrastructure. Cameras installed in cities, roads, banks, campuses, industries and homes record continuously, day and night, to support public safety, traffic management, access control and forensic investigation. With the rapid growth in the number of cameras and the steady increase in resolution and frame rate, the volume of video data generated by surveillance networks has grown enormously. Storing, transmitting and managing this data requires large storage capacity and high network bandwidth, which increases the cost and energy consumption of surveillance systems.

Most surveillance footage carries very little useful information. Cameras are usually fixed, so the background remains almost unchanged for long periods, and meaningful events such as the movement of persons or vehicles occupy only a small portion of each frame. Conventional video codecs such as H.264/AVC and H.265/HEVC reduce temporal redundancy through block-based motion estimation, but they apply the same block partitioning, search range and quality target to the whole frame. As a result, bits and computation are spent on regions that do not matter, while the objects that matter most receive no special treatment.

Recent advances in deep learning have produced object detectors that are both accurate and fast enough for real-time use. D-FINE is one such detector. It refines the localization of bounding boxes through fine-grained distribution refinement and achieves high accuracy at real-time speed. In this project, D-FINE is used to locate persons, vehicles and other objects in surveillance frames, and the detections are converted into a Region of Interest (ROI) mask. The mask guides an adaptive motion estimation stage in which object regions are processed with small blocks and a wide search range, while background regions use large blocks and a narrow search range. In this way, the proposed framework reduces storage and bandwidth requirements without sacrificing the visual quality of the regions that matter.

1.2. EXISTING WORK

Conventional surveillance systems store video using general-purpose codecs such as H.264 and H.265, together with fast block-matching search methods. These tools are mature and widely supported, but they are designed for generic video and use no knowledge of what appears in the scene.

- H.264/AVC and H.265/HEVC are the most common codecs in surveillance cameras and recorders because of their high compression efficiency.
- These codecs reduce temporal redundancy with block-based motion estimation, in which each block is matched against a reference frame within a fixed search range.
- Fast search algorithms such as the diamond search reduce the number of search points and the encoding time.
- Newer standards such as VVC give further bitrate reduction, but with higher encoding complexity.
- Recent surveillance-specific work uses object detectors such as Mask R-CNN and YOLO to remove irrelevant frames from long recordings.
- ROI-based coding and learned video codecs such as DVC have also been explored, but they are not combined with a real-time DETR-based detector to control block-level motion estimation.
- In this project, an existing YOLOv8-based compression method is used as the reference method, and its results are compared with those of the proposed D-FINE based method on the same test videos.

1.3. DRAWBACKS FOUND IN EXISTING WORK

Existing surveillance compression methods have several limitations that affect storage, bandwidth and image quality. Standard codecs treat all regions of a frame in the same way, so a large share of the bit budget is spent on static background. The motion search uses fixed parameters and is not linked to the content, which wastes computation on blocks that contain no motion. Methods that use detectors decide only which frames to keep, and inside the kept frames the objects and the background are still compressed alike. When the bitrate is reduced to save storage, faces, number plates and other fine details of moving objects may lose clarity. In the experiments of this work, the existing YOLOv8-based method also gave an unreliable storage saving: it reduced the size of only one of the four test videos and increased the size of the other three. Therefore, there is a need for a content-aware method that controls compression inside each frame.

Key points:

- Uniform treatment of object and background regions in standard codecs.
- Fixed block size and search range for the whole frame, causing redundant computation.
- Frame-level selection only, with no region-level control inside the retained frames.
- Loss of detail in important regions at low bitrates.
- High storage and bandwidth cost for continuous high-resolution recording.
- Inconsistent size reduction, with the output file becoming larger than the input for some videos.

1.4. NEED FOR THE PROPOSED WORK

The growing scale of surveillance networks makes efficient video compression a practical necessity. Because of the limitations of existing methods, there is a need for compression that understands the scene and directs bits and computation to the regions that carry useful information. The proposed work combines the D-FINE object detector with adaptive motion estimation to meet this need. D-FINE gives accurate ROI boxes at real-time speed, and the ROI mask controls the block size and search range of the motion estimation.

- Reduces storage and bandwidth by compressing the large static background more strongly.
- Preserves the visual quality of persons and vehicles using fine blocks and a wide search range.
- Lowers motion estimation computation by using large blocks and a narrow search range for the background.
- Uses an accurate, real-time detector so that the detection stage does not become a bottleneck.
- Suits large camera networks and edge devices with limited resources.
- Supports longer retention of surveillance footage at lower cost.

Objectives of the project:

- To detect objects of interest such as persons and vehicles in surveillance video using D-FINE.
- To generate a Region of Interest (ROI) mask from the detected bounding boxes for every frame.
- To apply adaptive motion estimation in which block size and search range depend on the ROI mask.
- To reduce file size while maintaining the visual quality of object regions.
- To evaluate the system using size reduction, PSNR and SSIM, and to compare it with the existing YOLOv8-based method.

CHAPTER-2
LITERATURE SURVEY

2.1. LITERATURE SURVEY

The demand for efficient storage and transmission of surveillance video has led to extensive research on video coding, motion estimation, ROI-based compression and object detection. Video coding standards have steadily improved compression efficiency, fast motion search methods have reduced encoding effort, and transformer-based detectors have made accurate real-time detection possible. Recent studies also explore the use of detection and learning to make compression content-aware. This literature review examines the work most closely related to the present project.

1. **"Overview of the H.264/AVC Video Coding Standard" (2003)**

   T. Wiegand, G. J. Sullivan, G. Bjontegaard, A. Luthra, IEEE Trans. Circuits Syst. Video Technol. [1]

   This paper describes H.264/AVC, which combines block-based motion estimation and compensation with transform, quantization and entropy coding. Variable block sizes and multiple reference frames give much higher compression efficiency than earlier codecs. The motion search and quality settings are applied uniformly and do not depend on the objects in the scene.

2. **"Overview of the High Efficiency Video Coding (HEVC) Standard" (2012)**

   G. J. Sullivan, J.-R. Ohm, W.-J. Han, T. Wiegand, IEEE Trans. Circuits Syst. Video Technol. [2]

   HEVC introduces coding tree units with flexible quadtree partitioning and improved motion vector prediction. It achieves about 50% lower bitrate than H.264 for similar quality, at the cost of much higher encoding complexity. All regions of a frame are still governed by the same rate-distortion rules.

3. **"Overview of the Versatile Video Coding (VVC) Standard and its Applications" (2021)**

   B. Bross et al., IEEE Trans. Circuits Syst. Video Technol. [3]

   VVC (H.266) adds more flexible partitioning and an extended set of prediction tools to obtain further bitrate reduction, especially for high-resolution content.

Encoder complexity increases again, and the standard does not use semantic information about the scene.

4. **“A New Diamond Search Algorithm for Fast Block-Matching Motion Estimation” (2000)**

   S. Zhu, K.-K. Ma, IEEE Trans. Image Process. [4]

   The diamond search uses a large and a small diamond-shaped pattern to find the best matching block with far fewer search points than a full search. The pattern is the same for every block and does not consider whether a block belongs to an object or to the background.

5. **“Motion-Vector-Driven Lightweight ROI Tracking for Real-Time Saliency-Guided Video Encoding” (2024)**

   T. Partanen, M. Kotajärvi, A. Mercat, J. Vanne, EUSIPCO [5]

   This work tracks the ROI with the motion vectors produced by the encoder, so that heavy detection does not have to run on every frame. For a single ROI and a 64-frame detection interval, it obtains up to 50-fold speedup with about 87% accuracy. It is demonstrated mainly for head tracking in video conferencing and does not adapt block-level motion search.

6. **“FRVC: Frame Relevance Based Video Compression for Surveillance Videos Using Deep Learning Methods” (2025)**

   Agrawal et al., PeerJ Computer Science [6]

   The method uses Mask R-CNN and YOLOv9 to find relevant frames of surveillance video and removes similar frames to build the compressed video. It reaches compression of up to 96.3%, 99.9% and 63.3% in three scenarios while keeping the original resolution and frame rate. It works at frame level, so inside retained frames the objects and background are encoded alike.

7. **“DVC: An End-to-End Deep Video Compression Framework” (2019)**

   G. Lu, W. Ouyang, D. Xu, X. Zhang, C. Cai, Z. Gao, IEEE/CVF CVPR [7]

   DVC replaces the main components of a hybrid codec with neural networks, using learned optical flow and auto-encoders for motion and residual, trained jointly. It outperforms H.264 in PSNR and is comparable to H.265 in MS-SSIM. It is general-purpose and requires training and heavy computation.

8. **“End-to-End Object Detection with Transformers (DETR)” (2020)**
   N. Carion et al., European Conference on Computer Vision (ECCV) [8]
   DETR treats detection as a direct set prediction problem with a transformer encoder-decoder and object queries. It removes anchors and non-maximum suppression, but it converges slowly and is computationally heavy.

9. **“DETRs Beat YOLOs on Real-time Object Detection (RT-DETR)” (2024)**
   Y. Zhao et al., IEEE/CVF CVPR [9]
   RT-DETR introduces an efficient hybrid encoder that makes DETR-style detection run in real time, matching or exceeding YOLO models in speed and accuracy without non-maximum suppression. It established the real-time DETR family on which D-FINE is built.

10. **“D-FINE: Redefine Regression Task in DETRs as Fine-grained Distribution Refinement” (2024)**
    Y. Peng et al., arXiv:2410.13842 [10]
    D-FINE predicts a probability distribution for each box edge and refines it layer by layer (FDR), and transfers knowledge from the last decoder layer to earlier layers (GO-LSD). It gives accurate localization at real-time speed. It is a detector only and has not been integrated with video compression.

Research gap: In the work reviewed here, a state-of-the-art real-time detector such as D-FINE is not used to drive block-level adaptive motion estimation for surveillance video. Standard codecs and fast search methods are not object-aware, and frame-level methods do not control compression inside a frame. The present project fills this gap by combining D-FINE based ROI detection with adaptive motion estimation.

CHAPTER-3
PROJECT DESCRIPTION

3.1. INTRODUCTION

Surveillance cameras record continuously, so the volume of stored and transmitted video is very large. Conventional codecs treat every region of a frame in the same way, although only a small part of the scene, such as a person or a vehicle, carries useful information. This project presents an intelligent framework that compresses surveillance video by combining object detection with adaptive motion estimation.

The D-FINE detector finds persons, vehicles and other objects in every frame, and the detections are converted into a Region of Interest (ROI) mask. The mask controls the block size and the search range used in motion estimation, so that object regions are encoded with fine detail and background regions are compressed more strongly. The system is implemented in Python using PyTorch, OpenCV and FFmpeg and is evaluated on four surveillance videos, with an existing YOLOv8-based method used for comparison.

3.2. BLOCK DIAGRAM

Fig.3.1 shows the block diagram of the proposed system. The surveillance video is read and split into frames. Each frame is passed to the D-FINE detector, which returns bounding boxes, class labels and confidence scores. The boxes are converted into an ROI mask, and the adaptive motion estimation stage uses the mask to select the block size and the search range for every block. The motion vectors and residuals are then entropy coded to produce the compressed video, which is finally evaluated with objective quality and compression metrics.

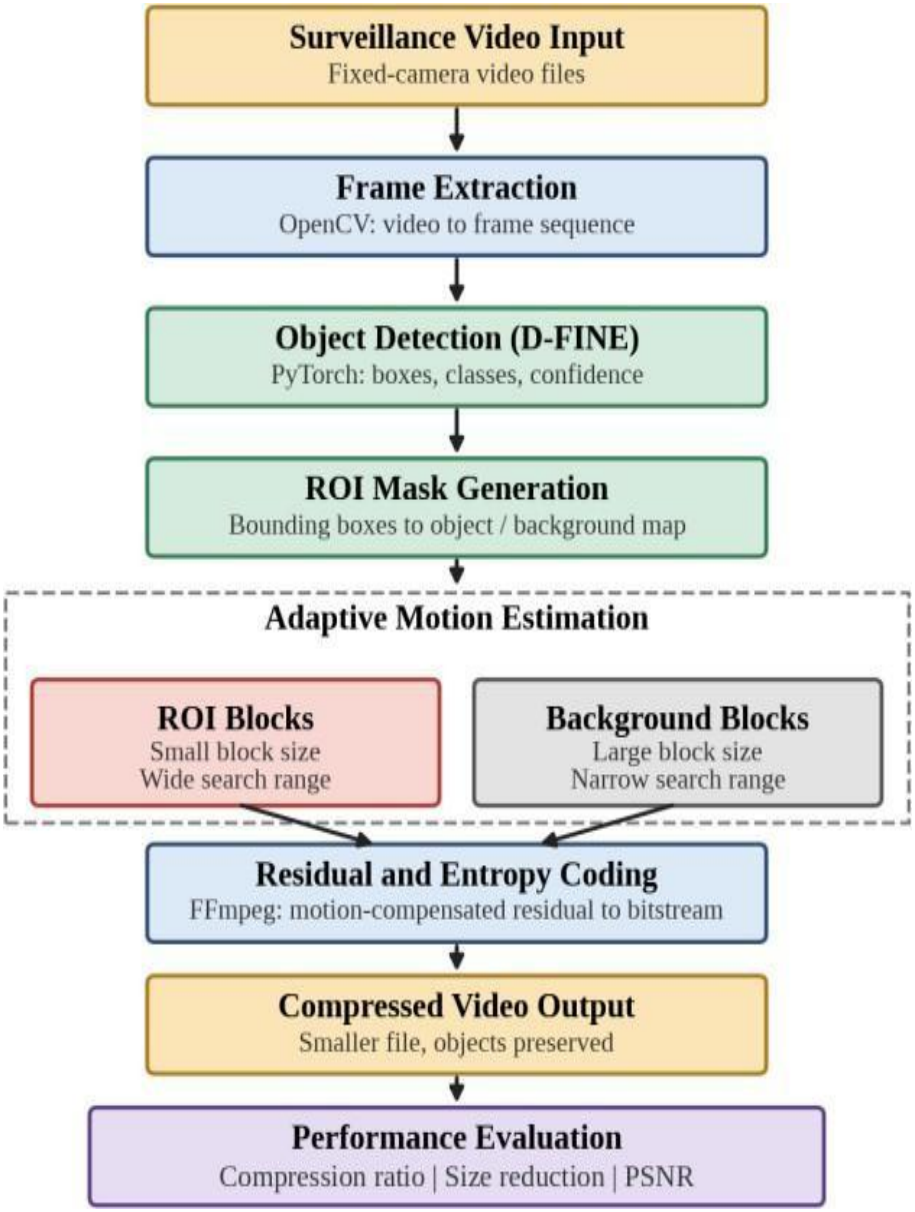


Fig.3.1 BLOCK DIAGRAM OF THE PROPOSED SYSTEM

3.3. WORKFLOW DIAGRAM

Fig.3.2 shows the decision made for every block in the adaptive motion estimation stage. The position of the block is checked against the ROI mask. A block inside the ROI is processed with a small block size and a wide search range, and a block outside the ROI is processed with a large block size and a narrow search range. In both cases the best match is found by block matching, and the resulting motion vector and residual are passed to the encoder.

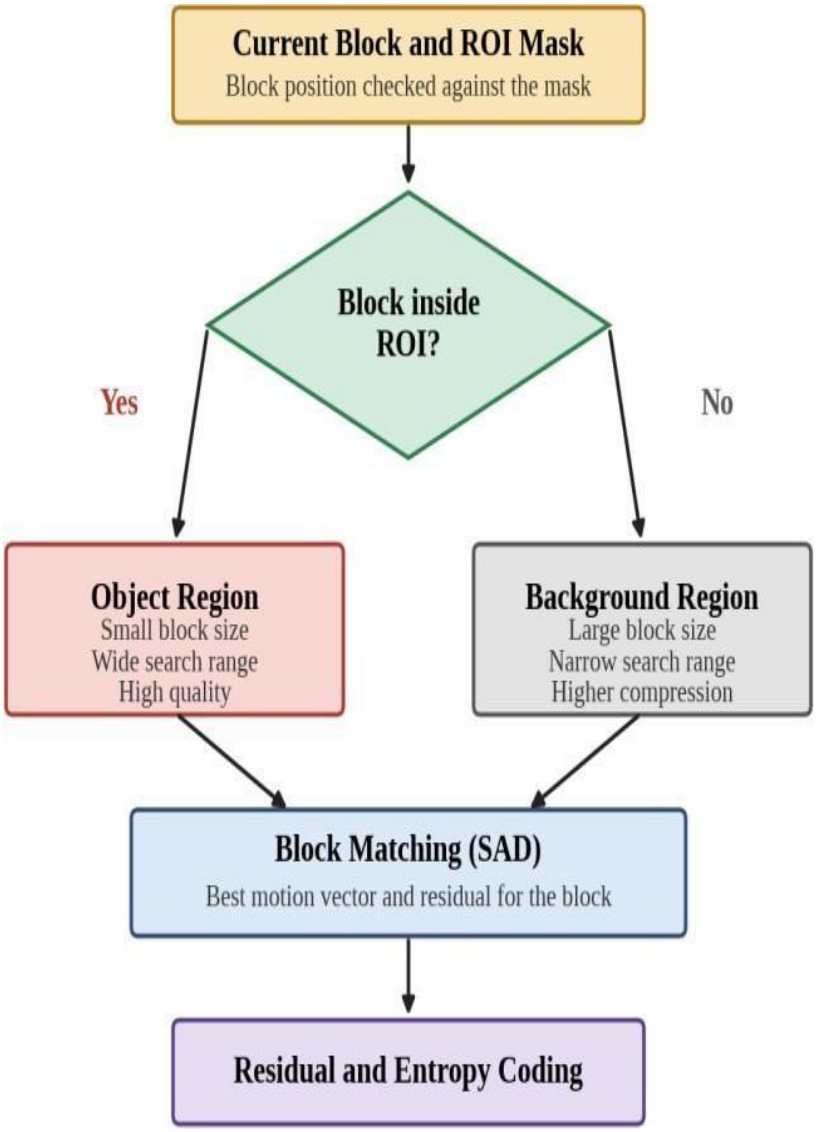


Fig.3.2 WORKFLOW OF THE ADAPTIVE MOTION ESTIMATION

3.4. EXPLANATION

3.4.1. Video Input and Frame Extraction

The process begins with the surveillance video recorded by a fixed camera. The video is opened with OpenCV and read frame by frame, and the frame rate, resolution and duration are noted for later comparison. Working on a frame sequence allows each frame to be analyzed and encoded with reference to its neighbours.

3.4.2. Object Detection using D-FINE

Every frame is given to a pretrained D-FINE model running in PyTorch. The model outputs bounding boxes with class labels and confidence scores. Detections below a confidence threshold are rejected, and only the classes relevant to surveillance, such as persons and vehicles, are retained. D-FINE is a real-time detector based on the DETR architecture. It improves box regression with two techniques:

- **Fine-grained Distribution Refinement (FDR):** each box edge is represented as a probability distribution over possible offsets, which is refined layer by layer in the decoder for finer localization.
- **Global Optimal Localization Self-Distillation (GO-LSD):** the localization knowledge of the final decoder layer is distilled into earlier layers during training, which improves accuracy without extra cost at inference.

D-FINE was chosen because its real-time speed prevents detection from becoming a bottleneck, its precise localization keeps the ROI small and accurate, and, being DETR-based, it needs no non-maximum suppression.

3.4.3. ROI Mask Generation

The retained bounding boxes are converted into a binary mask with the same size as the frame. Pixels inside any box are marked as ROI and all remaining pixels are marked as background. The mask is mapped to the block grid used by motion estimation, so that every block is labelled as an object block or a background block. This mask links the detector with the encoder.

3.4.4. Adaptive Motion Estimation

In block-based motion estimation, the current frame is divided into blocks and, for each block, the best matching block is searched within a window in the reference

frame. The matching error is usually measured as the Sum of Absolute Differences (SAD), and the displacement of the best match is the motion vector.

$$SAD(dx, dy) = \Sigma\Sigma \mid C(i, j) - R(i + dx, j + dy) \mid$$

Here C is the current block, R is the reference frame and (dx, dy) is a candidate displacement. A larger search range and a smaller block improve the prediction but increase computation and the number of motion vectors. The proposed method therefore adapts both parameters to the ROI label of the block, as listed in Table 3.1.

Region	Block Size	Search Range	Objective
Object (ROI) blocks	Small (fine) blocks	Wide	High accuracy and quality
Background blocks	Large (coarse) blocks	Narrow	Higher compression, lower computation

Table 3.1 Adaptive Parameters for ROI and Background Blocks

3.4.5. Residual Coding and Video Encoding

After motion compensation, the difference between the original block and its prediction, called the residual, is computed. The residual and the motion vectors are entropy coded, and the compressed video is written using FFmpeg. The settings used in the experiments are listed in Table 3.3.

3.4.6. Software Requirements

The complete pipeline is implemented in software. The tools used in the project are listed in Table 3.2.

Table 3.2 Software Requirements

Component	Details	Purpose
Programming language	Python	Implementation of the complete pipeline
Deep learning framework	PyTorch	Running the D-FINE model
Detection model	D-FINE (pretrained model)	Object detection and ROI generation
Image and video library	OpenCV	Frame reading, mask generation, visualization
Video encoding tool	FFmpeg (libx264)	Encoding of the compressed video

Table 3.3 Encoder Settings Used in the Experiments

Parameter	Value
Output codec	H.264 (libx264) using FFmpeg
Number of test videos	4
Existing method used for comparison	YOLOv8-based method
Proposed method	D-FINE based adaptive motion estimation
Size metrics	Compressed size (MB), size reduction (%)
Quality metrics	PSNR (dB), SSIM

3.4.7. Performance Metrics

The following metrics are used to evaluate the system. The compression ratio and the size reduction show how much smaller the video becomes after compression. A negative size reduction means that the output file is larger than the input file.

$$Compression\ Ratio = Original\ Size / Compressed\ Size$$

$$Size\ Reduction\ (\%) = (1 - Compressed\ Size / Original\ Size) \times 100$$

PSNR (Peak Signal-to-Noise Ratio) measures the pixel-level difference between the original and the reconstructed frames in decibels, and a higher value means a closer reconstruction.

$$PSNR = 10 \log10 (MAX^2 / MSE)$$

where MAX is the maximum pixel value and MSE is the mean squared error. In this work, the PSNR of each compressed video is computed with respect to the corresponding original video.

SSIM (Structural Similarity Index) measures the similarity between the original and the compressed frames in terms of luminance, contrast and structure. Its value lies between 0 and 1, and a value close to 1 means that the compressed frame is structurally almost identical to the original frame.

$$SSIM(x, y) = [(2\mu x\mu y + C1)(2\sigma xy + C2)] / [(\mu x^2 + \mu y^2 + C1)(\sigma x^2 + \sigma y^2 + C2)]$$

where μx and μy are the mean values, σx² and σy² are the variances and σxy is the covariance of the two images, and C1 and C2 are small constants that avoid division by zero.

3.4.8. Comparison with the Existing Method

To judge the benefit of the proposed method, the same four videos are also compressed with an existing YOLOv8-based method. The two methods are compared in terms of compressed size, size reduction and PSNR. SSIM is computed for the proposed method. The comparison results are presented in Chapter 4.

3.5. APPLICATIONS

The proposed D-FINE based adaptive compression system can be applied wherever fixed cameras record long videos in which only a small part of the scene is important.

1. **Public Safety and Smart Cities** – Reduces the storage and transmission load of city-wide camera networks while keeping persons and vehicles clear.
2. **Traffic Monitoring** – Preserves vehicle details for counting, classification and incident review on roads and highways.
3. **Banks and ATMs** – Compresses mostly empty indoor scenes strongly and keeps persons well preserved when they appear.
4. **Campus and Office Security** – Allows footage to be retained for longer periods at lower storage cost.
5. **Industrial Monitoring** – Records long operations in factories and warehouses with efficient storage.
6. **Edge Devices and Embedded Cameras** – Reduces the data to be uploaded from cameras with limited bandwidth.
7. **Cloud Video Archival** – Lowers cloud storage cost for long-term evidence.
8. **Forensic Investigation** – Keeps important regions sharp for identification of persons and vehicles.

3.6. ADVANTAGES

1. **Reduced Storage** – Strong compression of the static background gives a much smaller file.
2. **Lower Bandwidth** – Smaller video needs less network capacity for transmission.
3. **Preserved Object Quality** – Fine blocks and a wide search keep persons and vehicles sharp.
4. **Reduced Computation** – Large blocks and a narrow search for the background reduce motion estimation effort.
5. **Accurate ROI** – D-FINE provides precise bounding boxes at real-time speed.
6. **Content-Aware Design** – Bits are spent where the information is useful.
7. **Consistent Performance** – The size reduction stays between 80% and 88% for all test videos.
8. **Software-Based Solution** – Implemented with Python, PyTorch, OpenCV and FFmpeg without special hardware.
9. **Easy Integration** – Can be combined with existing encoding workflows and surveillance systems.

CHAPTER-4
RESULTS AND DISCUSSION

4.1. RESULTS

The proposed D-FINE based adaptive compression system was evaluated on four surveillance videos and compared with the existing YOLOv8-based method. Each video was compressed by both methods, and the compressed video was compared with the original video in terms of file size, PSNR and SSIM. The details of the test videos are given in Table 4.1.

Table 4.1 Details of the Test Videos

Video	File Name	Original Size (MB)
Video 1	not_shoplifting1.mp4	0.9434
Video 2	not_shoplifting2.mp4	1.6120
Video 3	not_shoplifting3.mp4	1.3040
Video 4	not_shoplifting4.mp4	1.0636
Total		**4.9230**

Table 4.2 gives the original size, compressed size, compression ratio and size reduction of the existing and the proposed methods for each video. The proposed method gives a size reduction between 80.226% and 87.556% for all four videos, with a compression ratio between 5.06 and 8.04. The existing method reduces the size of Video 1 by only 5.05%, and for Videos 2, 3 and 4 the compressed file is larger than the original file, which gives negative size reductions of −31.38%, −68.99% and −40.26%.

Table 4.2 Compression Results of the Existing and Proposed Methods

Video	Method	Original Size (MB)	Compressed Size (MB)	Size Reduction (%)	Compression Ratio
not_shoplifting1.mp4	Existing (YOLOv8)	0.9434	0.8960	5.05	1.05
	Proposed (D-FINE)	0.9434	0.1174	**87.556**	**8.04**
not_shoplifting2.mp4	Existing (YOLOv8)	1.6120	2.1180	-31.38	0.76
	Proposed (D-FINE)	1.6120	0.3188	**80.226**	**5.06**
not_shoplifting3.mp4	Existing (YOLOv8)	1.3040	2.2040	-68.99	0.59
	Proposed (D-FINE)	1.3040	0.2293	**87.556**	**5.69**
not_shoplifting4.mp4	Existing (YOLOv8)	1.0636	1.4920	-40.26	0.71
	Proposed (D-FINE)	1.0636	0.1987	**81.316**	**5.35**

The visual quality of the compressed videos is measured with PSNR and SSIM with respect to the original videos. The PSNR of the proposed method is higher than that of the existing method for every video, with a gain between 0.80 dB and 1.64 dB and an average gain of 1.19 dB. The SSIM of the proposed method is above 0.96 for all videos. The values are given in Table 4.3.

Table 4.3 Quality Metrics of the Compressed Videos

Video	Method	PSNR (dB)	SSIM
not_shoplifting1.mp4	Existing (YOLOv8)	35.1620	N/A
	Proposed (D-FINE)	**36.7985**	**0.963371**
not_shoplifting2.mp4	Existing (YOLOv8)	31.9720	N/A
	Proposed (D-FINE)	**32.7761**	**0.966796**
not_shoplifting3.mp4	Existing (YOLOv8)	32.3320	N/A
	Proposed (D-FINE)	**33.8115**	**0.967573**
not_shoplifting4.mp4	Existing (YOLOv8)	33.7350	N/A
	Proposed (D-FINE)	**34.5551**	**0.962830**
Average	**Existing (YOLOv8)**	**33.30**	**N/A**
	Proposed (D-FINE)	**34.49**	**0.9651**

Table 4.4 summarizes the overall result for the four test videos. The total size of the four original videos is 4.9230 MB. After compression with the existing method the total size becomes 6.7100 MB, which is an increase of 36.30%, whereas the proposed method reduces the total size to 0.8642 MB, which is an overall size reduction of 82.45%. The average of the individual size reductions of the proposed method is 82.879%.

Table 4.4 Overall Comparison Summary

Parameter	Existing (YOLOv8)	Proposed (D-FINE)
Number of videos	4	4
Total original size (MB)	4.9230	4.9230
Total compressed size (MB)	6.7100	0.8642
Overall size reduction (%)	-36.30	82.45
Average size reduction (%)	-33.90	82.879
Overall compression ratio	0.73	5.70
Average PSNR (dB)	33.30	34.49
Average SSIM	N/A	0.9651

The output figures below show the intermediate outputs of the system and the final comparison results. Fig.4.1 shows the motion vectors and Fig.4.2 shows the D-FINE detection output for a sample frame of the surveillance video. Fig.4.3 to Fig.4.6 present the comparison of the existing and the proposed methods.

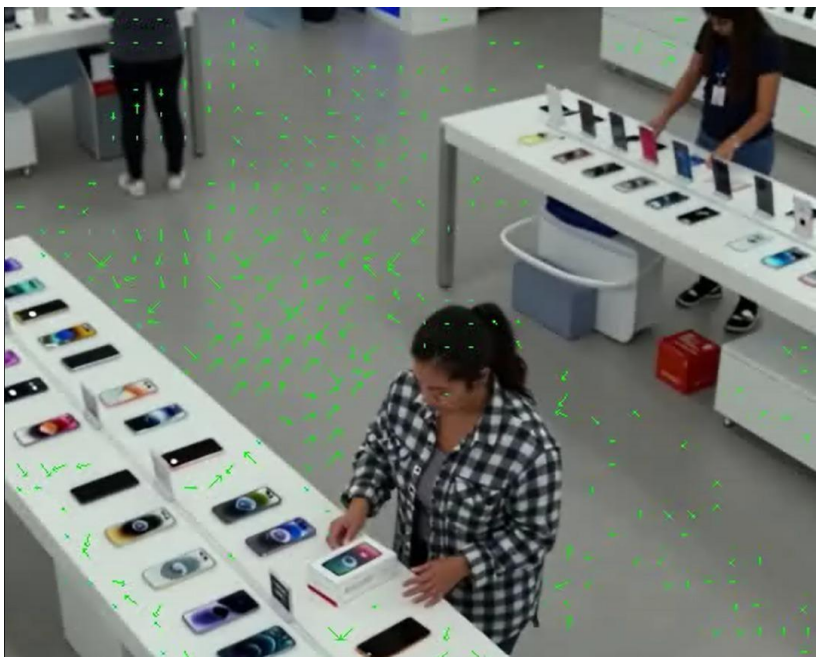


Fig.4.1 MOTION VECTOR OUTPUT FOR A SAMPLE FRAME

Fig.4.1 shows the motion vectors estimated for a sample frame of the surveillance video. Each green arrow represents the displacement of a block between consecutive frames, and its length and direction indicate the magnitude and direction of the motion. The vectors are concentrated in the central region of the scene around the moving person, while large static areas show fewer or shorter vectors. The encoder uses these vectors to predict the current frame from the reference frame, so that only the residual has to be coded.



Fig.4.2 D-FINE DETECTION OUTPUT FOR A SAMPLE FRAME

Fig.4.2 shows the output of the D-FINE detector for a sample frame. The red rectangles are the bounding boxes predicted for the persons and for the mobile phones displayed on the table. These boxes are converted into the ROI mask, so that the detected regions are treated as object blocks with a small block size and a wide search range, and the remaining area is treated as background with a large block size and a narrow search range.

Video	Method	Original Size (MB)	Compressed Size (MB)	Size Reduction (%)	Compression Ratio
not_shoplifting1.mp4	Existing (YOLOv8)	0.9434	0.8960	5.05	1.05
	Proposed (D-FINE)	0.9434	0.1174	**87.556**	**8.04**
not_shoplifting2.mp4	Existing (YOLOv8)	1.6120	2.1180	-31.38	0.76
	Proposed (D-FINE)	1.6120	0.3188	**80.226**	**5.06**
not_shoplifting3.mp4	**Existing (YOLOv8)**	1.3040	2.2040	-68.99	0.59
	Proposed (D-FINE)	1.3040	0.2293	87.556	5.69
not_shoplifting4.mp4	**Existing (YOLOv8)**	1.0636	1.4920	-40.26	0.71
	Proposed (D-FINE)	1.0636	0.1987	81.316	5.35

Fig.4.3 COMBINED COMPARISON TABLE OF EXISTING YOLOV8 AND PROPOSED D-FINE

Fig.4.3 shows the combined comparison table generated for the four videos. For every video it lists the original size, compressed size, size reduction, PSNR and SSIM

of the existing and the proposed methods. The values are the same as those given in Table 4.2 and Table 4.3.

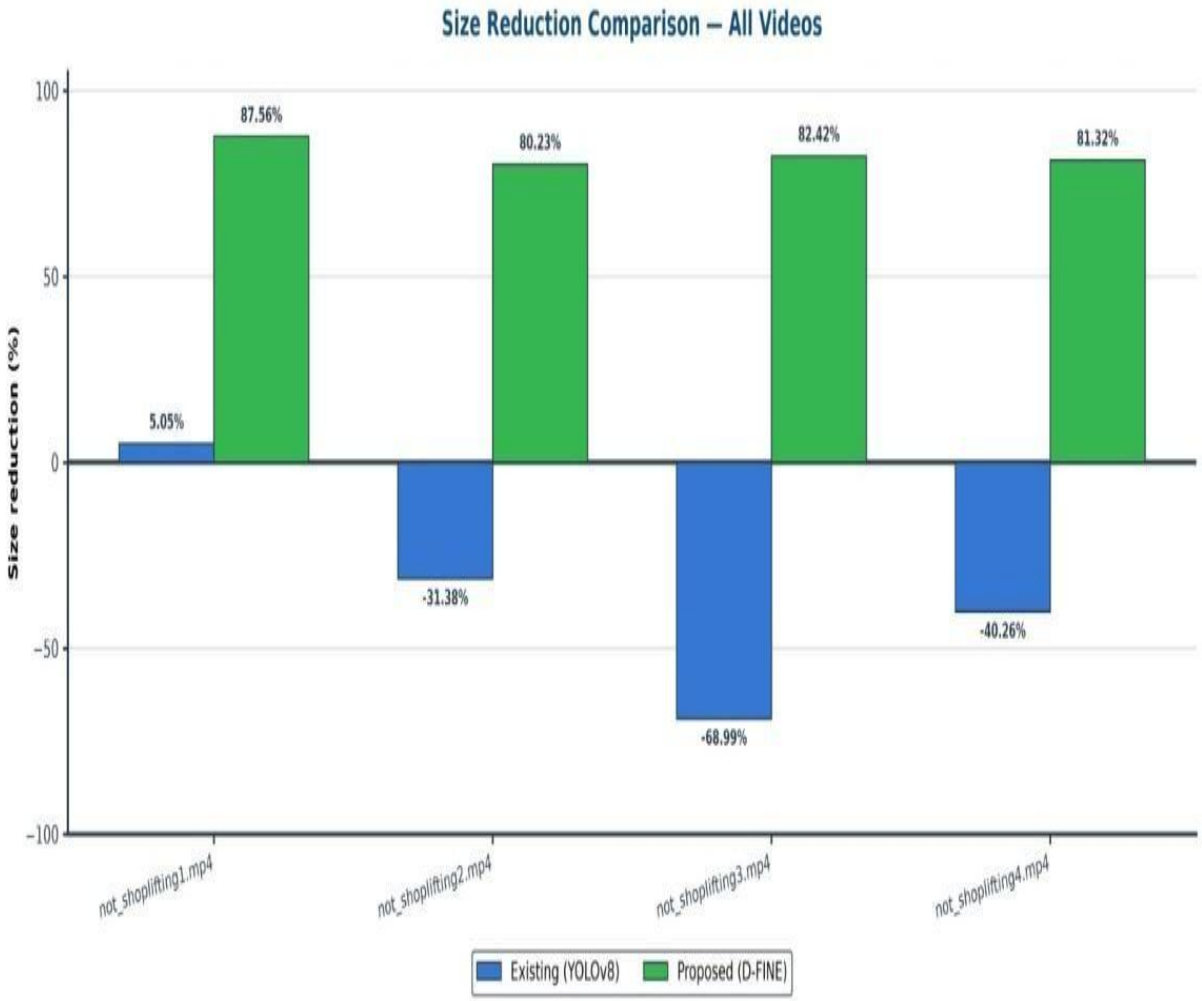


Fig.4.4 SIZE REDUCTION COMPARISON FOR ALL VIDEOS

Fig.4.4 compares the size reduction of the two methods. The proposed D-FINE method (green bars) gives a high positive size reduction of 87.56%, 80.23%, 82.42% and 81.32% for the four videos. The existing YOLOv8 method (blue bars) gives only 5.05% for Video 1, and the bars of Videos 2, 3 and 4 lie below zero, which means that the output file is larger than the original file. The proposed method therefore saves storage for every video, whereas the existing method does not.



Fig.4.5 PSNR COMPARISON FOR ALL VIDEOS

Fig.4.5 compares the PSNR of the two methods. The proposed method gives a higher PSNR than the existing method for all four videos: 36.80 dB against 35.16 dB for Video 1, 32.78 dB against 31.97 dB for Video 2, 33.81 dB against 32.33 dB for Video 3 and 34.56 dB against 33.74 dB for Video 4. The average PSNR is 34.49 dB for the proposed method and 33.30 dB for the existing method. The proposed method thus gives much smaller files and also a better reconstruction quality.

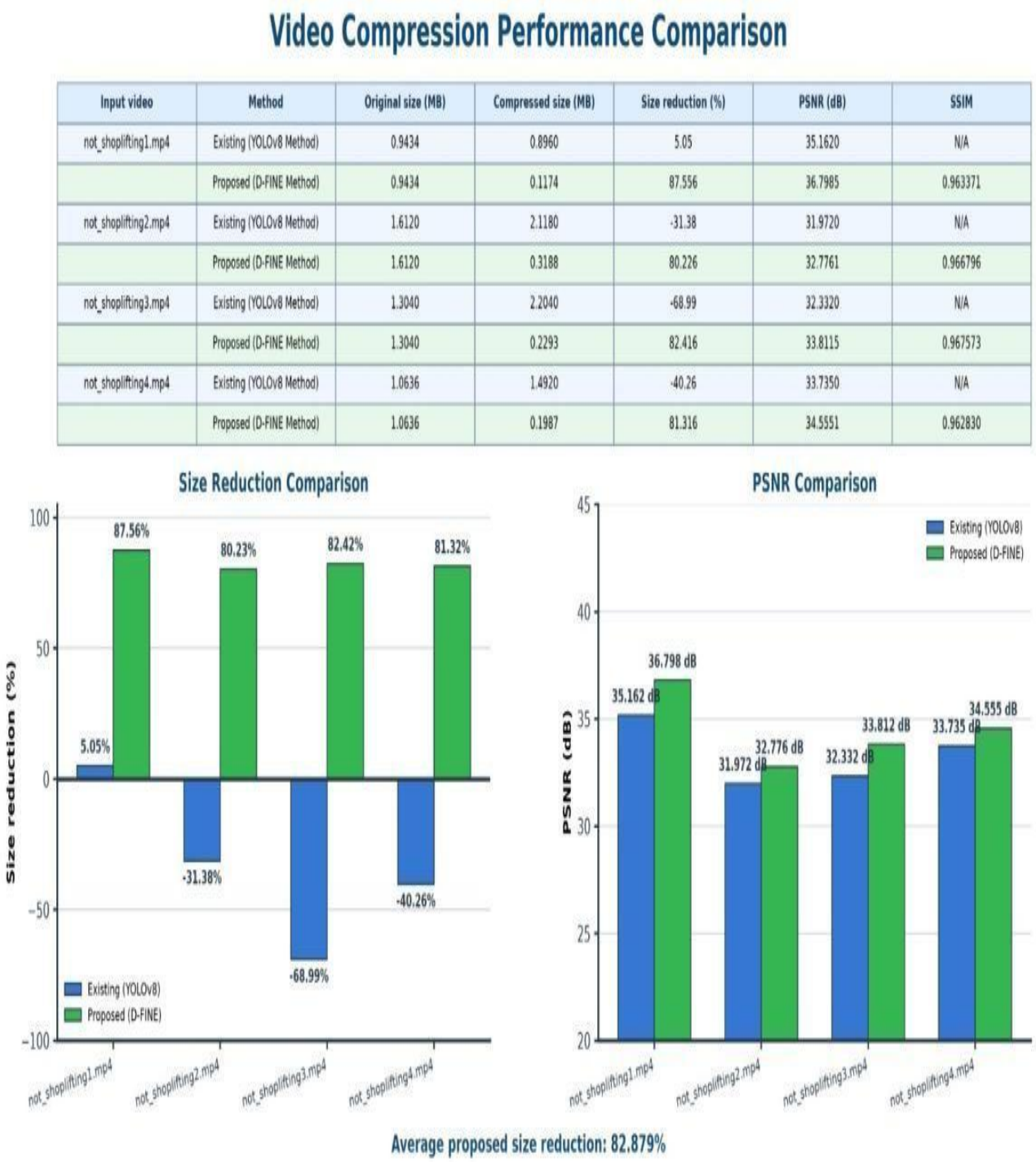


Fig.4.6 OVERALL PERFORMANCE COMPARISON OF THE EXISTING AND PROPOSED METHODS

Fig.4.6 gives the overall performance comparison in a single view, with the result table, the size reduction chart and the PSNR chart. The average size reduction of the proposed method is 82.879%.

4.2. DISCUSSION

The proposed system combines D-FINE object detection with adaptive motion estimation to compress surveillance video in a content-aware way. D-FINE locates persons and other objects accurately, and the resulting ROI mask allows the encoder to use fine blocks and a wide search range for the objects and coarse blocks and a narrow search range for the background. The comparison with the existing YOLOv8-based method on four test videos shows that the proposed method gives a much better storage saving and also a better quality.

The existing method does not give a dependable saving. It reduces the size of Video 1 by only 5.05%, and for the other three videos the compressed file grows by 31.38%, 68.99% and 40.26%. The proposed method reduces the size of every video by more than 80%. Video 1 gave the highest size reduction of 87.556% and Video 2 the lowest of 80.226%, so the spread between the best and the worst video is only about 7 percentage points. Video 1 also gave the highest PSNR (36.80 dB) and Video 2 the lowest (32.78 dB), and the same order is seen for the existing method, which shows that the content of the video affects the quality of both methods.

In terms of quality, the PSNR of the proposed method is higher than that of the existing method for all four videos, with an average gain of 1.19 dB. The PSNR values of 32.78 dB to 36.80 dB, together with the SSIM values between 0.9628 and 0.9676, show that the compressed videos remain structurally very close to the original videos even at a size reduction above 80%. In total, the four videos occupy 4.9230 MB originally, 6.7100 MB after the existing method and only 0.8642 MB after the proposed method, which saves about 4.06 MB. The motion vector output in Fig.4.1 and the detection output in Fig.4.2 confirm that the system identifies the moving persons and objects in the scene, which are the regions that must be preserved.

The method also has limitations. Its performance depends on the accuracy of the detector, so an object that is missed is treated as background and may lose some quality. The approach is designed for fixed-camera surveillance, in which the background is largely static, and it may give smaller gains for moving cameras. SSIM was computed only for the proposed method, so the structural similarity of the two methods could not be compared directly. A lower confidence threshold and simple motion cues can reduce the effect of missed detections.

CHAPTER-5
CONCLUSION & FUTURE SCOPE

5.1. CONCLUSION

The D-FINE based intelligent surveillance video compression system presents an efficient and content-aware solution to the storage and bandwidth problem of continuous video surveillance. Conventional codecs treat all regions of a frame in the same way and spend a large part of their bit budget on static background. To overcome this, the project uses the D-FINE object detector to identify the important objects and to generate an ROI mask for every frame.

Guided by the mask, the adaptive motion estimation stage applies small blocks and a wide search range to object regions and large blocks and a narrow search range to background regions. This focuses bits and computation on the regions that carry useful information, reduces the file size and preserves the visual quality of persons and vehicles. The experiments on four surveillance videos show that the proposed method reduces the file size of every video by 80.23% to 87.56%, with an average size reduction of 82.879%, whereas the existing YOLOv8-based method reduces only one video by 5.05% and increases the size of the other three. The proposed method also gives a higher PSNR for all videos, with an average of 34.49 dB against 33.30 dB, and an average SSIM of 0.9651. These results confirm that object-aware compression can reduce storage requirements significantly without a loss of quality. The framework is implemented entirely in software using Python, PyTorch, OpenCV and FFmpeg, which makes it practical for surveillance systems without special hardware.

5.2. FUTURE SCOPE

The proposed system can be extended using RF-DETR to improve its performance and prepare it for real deployment.

1. **Edge Deployment:** The system can run on edge devices and embedded cameras so that compression takes place close to the source and upload bandwidth is reduced.
2. **Multi-Camera and Live Streaming:** Support for several cameras and live streams can be added, with detection and encoding running in parallel for real-time operation.
3. **Class-Based Priority:** Persons can be given a higher quality level than vehicles or other objects.
4. **ROI Tracking:** Object tracking or motion-vector based ROI tracking can avoid running the detector on every frame and reduce the detection overhead.
5. **Integration with Modern Codecs:** The method can be combined with HEVC, VVC or learning-based codecs for higher compression efficiency.
6. **Wider Testing:** Larger and more diverse datasets, including night-time and low-light videos, can be used to test robustness. The evaluation can also be extended with bitrate, encoding time and SSIM of the existing method.

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